AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I165*

Team Name: *Yeezy*

Domain: *Domain - 7 : Digital Technologies, AI and Cybersecurity*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals

Problem Statement

Many students struggle to understand complex concepts and receive individual attention. Therefore, there is a need for an AI tutor that provides personalised explanations, instant feedback, and interactive learning support."

Problem Understanding

- 1. Who experiences this problem?
  - Engineering students who struggle with Maths and Physics.
  - Students who need personal attention from teachers.
  - Students who feel shy to ask questions.
  - Students who get distracted during online classes

- 2. What causes the problem?
  - Difficult concepts are not always easy to understand.
  - Teachers cannot give equal attention to every student.
  - Students do not always get instant feedback.
  - Every student learns at a different speed.
  - Online classes can be distracting.

- 3. How does it affect students and teachers?
  - Students may not understand concepts clearly.
  - They may repeat the same mistakes.
  - Students may lose confidence and interest in learning.
  - Marks may decrease due to poor understanding.
  - Teachers may find it difficult to identify each student's weaknesses.

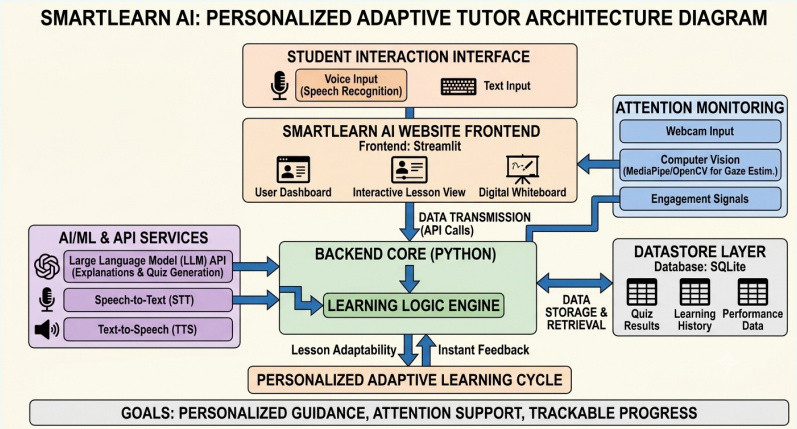
Problem Understanding

- Existing gaps challenges
  - Personalized learning: AI tutors may not explain topics in a way every student understands. - Finding weak areas: Students may repeat mistakes without knowing what to improve.
  - Staying focused: Students may get distracted during online classes.
  - Tracking progress: Students may not know how much they have improved.
  - Main challenge: Building an AI tutor that explains clearly, identifies weaknesses, tracks progress, and helps students stay focused.

Our Proposed Solution

- 1. Personalised AI tutor :
  SmartLearn AI works like a personal teacher for engineering mathematics and physics. it explains difficult topics in simple words, solves problems step by step, and allows students to ask questions through text or voice whenever they need help.
  2. Smart and adaptive learning :
  The system gives quizzes, checks students' answers, and identifies the topics they find difficult. Based on their performance, it provides simpler explanations and extra practice questions to help them improve their understanding.
  3. Attention monitoring and progress tracking :
  SmartLearn AI can remind students to pay attention when they appear distracted during online lessons. it also tracks their learning progress, remembers their previous performance, and suggests topics they need to revise to improve their results.

Technical Approach



Students interact with the SmartLearn AI website through text or voice input. The Python backend processes their questions and connects to an LLM API to generate personalized explanations and practice questions. The system analyzes quiz results to identify weak topics and uses SQLite to store learning history and performance data. Based on the results, the AI tutor adapts future lessons to match each student's learning needs and provides instant feedback and progress tracking. Optional computer vision-based attention monitoring can also send reminders to help students stay focused during learning.

Tech Stack

Frontend: Streamlit

Backend: Python

Database: SQLite

AI/ML, APIs, or Hardware: LLM API

Innovation & Uniqueness

- 1. What is new or different?
  Teaches in a cycle: Doesn't just give answers—it explains, gives quizzes, finds weak spots, and changes future lessons. Interactive learning: Students can talk, listen, and follow step-by-step solutions on a digital whiteboard. Focus support: Reminds students gently when they get distracted during lessons.
  2. Why does this work?
  Great for hard subjects: Breaks down complex Maths and Physics problems step by step. Keeps students active: Quizzes and reminders stop students from zoning out. Remembers mistakes: Saves past scores to help students review what they missed.
  3. Main Differentiator SmartLearn AI works like a personal tutor—it adapts lessons to your speed, notices when you lose focus, and helps you improve

Feasibility & Scalability

- **1. Feasibility:** We can build SmartLearn AI as a website using AI tools. It will teach engineering mathematics and physics, answer questions, explain problems step by step, and conduct quizzes.
- **2. Scalability:** We can expand the website by adding more subjects, languages, and features. It can also be improved to support more students in the future.
- **3. Risks:** The main challenges are AI costs, incorrect answers, internet issues, and difficulty in detecting distractions. We can reduce these problems by using affordable tools, checking answers, and giving students gentle reminders.

Cost for Product (Optional)

- **Development Cost:** 0 (Using free, open-source tools like Python, Streamlit, SQLite, and MediaPipe)
- **AI API Cost:** $\approx$ 800 – 1,500/month (For LLM and speech API usage during testing)
- **Hosting & Maintenance:** 0 (Free deployment using Streamlit Community Cloud)
- **Estimated Total:** $\approx$ 1,000/month for the prototype phase

Use Cases

- **Use Case 1: Preparing for Exams (Engineering Student)**
  A student struggling with Engineering Mathematics uses SmartLearn AI to learn step-by-step problem-solving. The AI explains formulas, gives practice quizzes, identifies weak topics, and provides targeted revision before exams.

- **Use Case 2: Staying Focused During Online Study (Distracted Learner)**
  A student gets frequently distracted during long online study sessions. SmartLearn AI detects signs of inattention via optional webcam monitoring and sends gentle reminders, helping the student refocus and complete their physics lesson.

- **Use Case 3: Interactive Voice Whiteboard Learning (Hands-Free Learner)**
  A student who prefers interactive learning asks questions aloud using voice input. SmartLearn AI speaks back the explanation while drawing step-by-step derivations on the digital whiteboard for better visual understanding.

Impacts and Benefits

- **Benefit to target users:** Empowers engineering students with personalized 24/7 support, step-by-step visual derivations, and interactive voice feedback.
- **Social, economic, or environmental impact:** Democratizes high-quality STEM education, making personalized tutoring accessible and affordable for all students.
- **Expected measurable outcome:** Improved quiz scores, higher subject retention, and increased learning focus during online study sessions.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 4	Target 4.4	Quality Education	Provides personalized 24/7 STEM tutoring, improving student learning outcomes and technical skill development.
SDG 9	Target 9.5	Industry, Innovation and Infrastructure	Integrates advanced AI technology to innovate and modernize digital education infrastructure.

Individual Member Contribution

Member	Contribution or responsibility
Dhanush.M , Naren.S.S	Research and problem statement formulation
M.Mohamed Navith , Surya.S	AI architecture design and UI/UX prototyping
Naren .S.S , Surya.S	Development, testing, and system integration
M.Mohamed Navith,Dhanush.M	Documentation, presentation, and slide creation

Yukti ID

IR2026-1051075